CLOSURES OF IRREDUCIBLE SETS AND THE UNION DESCRIPTION OF THE IRREDUCIBLE COMPONENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The definition of an irreducible topological space, of an irreducible component and of an irreducible subset. Set up in the author's section on irreducible components, immediately before the counted statement. Fixes the objects of the four numbered assertions.

Lemma 0.1. *Let X be a topological space.*

(1) *If $T \subset X$ is irreducible so is its closure in X.*
(2) *Any irreducible component of X is closed.*
(3) *Any irreducible subset of X is contained in an irreducible component of X.*
(4) *Every point of X is contained in some irreducible component of X, in other words, X is the union of its irreducible components.*

Proof. Let $\overline{T}$ be the closure of the irreducible subset T. If $\overline{T} = Z_1 \cup Z_2$ with $Z_i \subset \overline{T}$ closed, then $T = (T \cap Z_1) \cup (T \cap Z_2)$ and hence T equals one of the two, say $T = Z_1 \cap T$. Thus clearly $\overline{T} \subset Z_1$. This proves (1). Part (2) follows immediately from (1) and the definition of irreducible components.

Let $T \subset X$ be irreducible. Consider the set A of irreducible subsets $T \subset T' \subset X$. Note that A is nonempty since $T \in A$. There is a partial ordering on A coming from inclusion: $T' \leq T'' \Leftrightarrow T' \subset T''$. Choose a maximal totally ordered subset $A' \subset A$, and let $E = \bigcup_{T' \in A'} T'$. We claim that E is irreducible. Namely, suppose

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that $E = Z_1 \cup Z_2$ is a union of two closed subsets of E. For each $T' \in A'$ we have either $T' \subset Z_1$ or $T' \subset Z_2$, by irreducibility of T'. Suppose that for some $T'_0 \in A'$ we have $T'_0 \not\subset Z_1$ (say, if not we're done anyway). Then, since A' is totally ordered we see immediately that $T' \subset Z_2$ for all $T' \in A'$. Hence $E = Z_2$. This proves (3). Part (4) is an immediate consequence of (3) as a singleton space is irreducible. $\square$